HMM-GATED PROTEIN LANGUAGE MODEL EMBEDDINGS FOR BACTERIAL CULTIVATION CONDITION PREDICTION

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Abstract

The majority of microbial diversity remains uncultivated, in large part because we do not know what cultivation conditions a given lineage requires. Existing genome based phenotype predictors operate on a single feature family, protein family inventories, codon usage statistics, or single trait marker panels, and most binarize continuous phenotypes such as optimum temperature, pH, and salinity into thermophile/non thermophile labels. Here we introduce **HMM-gated protein language model embeddings (PTPE, phenotype-targeted PLM embeddings)**: for each genome we run pyhmmer against eight curated phenotype relevant HMM marker families (oxygen handling, thermo tolerance, pH homeostasis, osmotic response, vitamin biosynthesis, nitrogen cycling, carbon utilization, and a "special" category), embed only the matched proteins with ESM 2 t30, and mean pool **within each category** to produce a compact 8 × 641 dimensional functional fingerprint of the genome. We integrate PTPE with five additional feature paths, amino acid composition, MediaDive recipe metadata, Pfam HMM marker counts, KEGG module fractional completeness (570 modules from the completed KOfam scan), and parsed BacDive isolation metadata, for a total of 6,313 features per genome, and train a multi task XGBoost over 46,029 BacDive strains (22,300 unique genomes) for four cultivation targets (optimum temperature, pH, oxygen requirement, salt tolerance) with five fold family grouped cross validation. PTPE adds modest, target dependent lift over the five path baseline: optimum temperature MAE improves from 2.74 to 2.67 °C, a 2.4% reduction, salt MAE from 1.94 to 1.92%, a 1.1% reduction, and pH MAE from 0.473 to 0.469, a 1.0% reduction; oxygen F1 macro regresses from 0.412 to 0.402, a 2.4% decrease, suggesting that frozen mean pool does not unlock the full per marker PLM signal for classification tasks. We benchmark GenomeSPOT on a deterministic 5,000 unique genome subset of the same family heldout manifest; GenomeSPOT completes 5,000/5,000 genomes with no failures but is less accurate than our tabular/hybrid stack for optimum temperature (4.39 vs. 2.67 °C MAE) and pH (0.61 vs. 0.47 MAE), while salt is close (1.98 vs. 1.92% MAE). We then fine tune ESM 2 t12 with LoRA adapters on the HMM gated marker sequences. Fold 0 LoRA is worse than tabular XGBoost for temperature and pH, comparable for salt, but sharply improves oxygen classification (macro F1 0.945 vs. 0.402 for the five fold tabular mean). An oxygen only LoRA run does not improve further (macro F1 0.917), so the retained production system is a hybrid: tabular XGBoost for temperature, pH, and salt; all task LoRA for oxygen; and the tabular MediaDive recommender for medium ranking. Applied to 5,000 GTDB derived uncultured catalog genomes, the hybrid predictor assigns LoRA oxygen labels to 4,999 genomes and falls back to tabular oxygen for one missing marker genome, yielding 3,026 predicted aerobes and 1,974 predicted anaerobes.

Keywords microbial cultivation · phenotype prediction · protein language models · HMM profiles · KEGG modules · BacDive · uncultivated microorganisms

1. Introduction

Cultivation of microbes from environmental samples is bottlenecked not by sequencing or isolation hardware but by the prior question of **which conditions to use**: optimum temperature, pH, oxygen tolerance, and salinity must be chosen, typically from broad ranges, before a strain can be enriched in pure culture. For the >99% of bacterial and archaeal lineages that have never been cultivated [Steen 2019; Lloyd 2018], 16S placement provides only weak constraints on these conditions: phylogenetically close strains routinely differ by 10 °C in optimum temperature or several pH units. Direct prediction of cultivation conditions from genome sequence would lower this barrier substantially.

A growing literature attempts this. The PICA framework [Feldbauer 2015] introduced support vector classification on cluster of orthologous groups presence/absence for ten binary traits, establishing that 60 70% genome completeness is sufficient. Koblitz et al. [Koblitz 2025] scaled this approach to 21,168 BacDive type strains with random forests on Pfam annotations, achieving F1 in 0.85 0.95 on eight binary traits and integrating 50,396 predictions back into BacDive. Li et al. [Li 2023] showed that random forests on KEGG ortholog presence/absence predict carbon utilization traits at >90% within clade accuracy but fail on phylogenetically out of clade tests, exposing the dependency of presence/absence features on phylogenetic signal. Single trait specialist tools have been built for sporulation [SpoMAG 2025], growth rate [Oduwole 2025; gRodon], and culturability [Functional Genomic Signatures 2025]. Most recently, rule based explainable predictors [Máša 2025] have shown that knowledge graph derived organismal traits yield interpretable medium preference rules, albeit only on two cultivation media (Marine Broth, GYM Streptomyces).

A common gap across this literature is that **protein family inventories are coarse**. A Pfam hit is binary: "does this genome contain a cytochrome c oxidase?". It collapses the substantial *sequence level* variation between different cytochromes, variation that determines whether the organism is microaerophilic, strictly aerobic, or facultative. Protein language models (PLMs) such as ESM 2 [Lin 2023] are designed to expose precisely this continuous variation: two proteins with weak Pfam annotation agreement but high ESM 2 similarity occupy nearby points in embedding space. But naive whole proteome PLM pooling, averaging ESM 2 across every protein in a genome, drowns the few phenotype relevant proteins (12 cytochromes) in the ~4,000 housekeeping ones, producing a feature vector that is biologically dilute.

We propose a middle path: **use the HMM as a gate for which proteins to embed**. For each genome we run pyhmmer against a curated panel of 48 phenotype relevant Pfam HMMs grouped into 8 categories, embed only the matched proteins with ESM 2 t30, and mean pool *within* each category. The result is a phenotype targeted protein language model embedding (PTPE) of 8 × 641 dimensions per genome. PTPE keeps the *specificity* of HMM based feature selection and the *continuous functional resolution* of PLMs. We integrate PTPE with five additional feature paths and train a multi task XGBoost on 46,029 BacDive strains with family grouped cross validation, the largest BacDive anchored phenotype prediction corpus published to date (~2× the size of Koblitz 2025).

Our contributions are:

1. **PTPE construction (§2.1, §4.5).** A novel feature type for genome level phenotype prediction. To our knowledge no prior work in the eight surveyed BacDive era papers uses HMM gated PLM mean pooling.
2. **Multi source feature fusion at BacDive scale (§2.2).** 46,029 strains × 6,313 features integrating composition, MediaDive, Pfam markers, KEGG modules, isolation metadata, and PTPE. 5 fold family grouped CV with the strongest pre PTPE baseline released as a reproducible comparator.
3. **An honest empirical evaluation (§2.3).** PTPE adds modest, target dependent lift on regression targets (1 to 2.4%) but slightly regresses oxygen F1. We do not overclaim; we characterize where frozen PTPE helps and where it does not.
4. **A fold 0 LoRA result and deployed hybrid predictor (§2.4 2.6).** End to end trained ESM 2 adapters improve oxygen classification substantially but do not replace tabular heads for continuous targets. We therefore deploy a phenotype specific hybrid predictor and apply it to 5,000 uncultured catalog genomes.
5. **A GenomeSPOT comparator (§2.7).** We run GenomeSPOT on 5,000 held out BacDive derived genomes from the same family grouped manifest, producing an external condition trait comparator with no failed genomes.

2. Results

2.1 Phenotype targeted PLM embeddings (PTPE)

For each BacDive genome (~22,300 unique genome accessions across ~46,000 strains), we run pyrodigal to predict the proteome, then pyhmmer against a curated 48 marker panel grouped into 8 phenotype categories (Methods §4.4). The 48 marker panel was validated against the InterPro DESC field for each Pfam ID (`scripts/23_verify_markers.py`) to remove the small fraction of accessions that had been repurposed across Pfam versions. Marker categories and representative members (full panel in Table S1):

Category	n markers	Representative HMMs
temperature	6	Hsp70_DnaK, Cpn60_GroEL, CSD_cold_shock
pH	8	NhaA_Na_H_exch, ATP_synth_alphabeta, V_ATPase_subH_N
oxygen	12	COX1_aerobic, Catalase, SOD_FeMn, FeFe_hyd_anaerobic
salt	6	KdpD_osmosensor, BCCT_compatible, EctC_ectoine_synth
vitamin	6	TP_methylase_B12, FolB_folate, DHBP_riboflavin

Category	n markers	Representative HMMs
nitrogen	3	NifH_nitrogenase, NifDK_nitrogenase
carbon	5	RuBisCO_large_form1, Alpha_amylase, Cellulase_GH5
special	2	Molybdopterin_OR, UvrD_helicase_C

For each genome, the proteins matching any HMM in category *c* are encoded by frozen ESM 2 t30 (640 dim mean pooled across residues), then averaged within the category to yield one 640 dim category vector plus one integer count of contributing proteins. Concatenated across the 8 categories this produces a 5,128 dimensional PTPE feature vector per genome (8×641: 640 embedding + 1 count). Median hit counts per category are 6 (nitrogen) to 16 (oxygen, capped at 16 in extraction). 5,128 dimensions makes PTPE the single largest feature path in our model, complementing 355 composition columns, 570 KEGG module columns, 144 Pfam marker columns, 65 isolation metadata one hot columns, and 5 MediaDive recipe columns.

2.2 Multi source feature fusion at BacDive scale

We trained a multi task XGBoost model (Methods §4.6) on 46,029 BacDive strains for four cultivation targets, optimum temperature (°C, regression), optimum pH (regression), oxygen requirement (4 class classification: aerobe / facultative anaerobe / microaerophile / anaerobe), and salt tolerance (% NaCl, regression). The training table joins six feature paths on `genome_accession` and `bacdive_id`:

Feature path	n columns	Source
Composition / codon / tetranucleotide	355	pyrodigal CDS → amino acid + codon + 4 mer stats
MediaDive recipe aggregates	5	medium pH/NaCl/n_media this strain grows on
Pfam HMM markers (curated 48)	144	bit score + presence per marker × 3 stats
KEGG module completeness	570	full KOfam scan + module rule evaluator
Isolation metadata (numeric + one hot)	111	parsed BacDive JSON: lat/lon/continent/host kingdom
Phenotype targeted ESM 2 (PTPE)	**5,128**	**HMM gated mean pool per category, this work**
Total	**6,313**	

Cross validation uses 5 fold GroupKFold over BacDive's `family` field, with genus and then species fallbacks for the 4.5% of strains lacking family assignment. This prevents trivial phylogenetic leakage: every family appears in exactly one fold's test set.

2.3 PTPE adds modest, target dependent lift

We trained the full fusion model with and without the PTPE feature path on the same 5 fold splits, keeping all other features and hyperparameters identical. Table 1 reports per target 5 fold CV means (full per fold values in Table S2; raw results in `artifacts/baseline_results.json` and `artifacts/baseline_results_pre_pme.json`).

Table 1. Cultivation condition prediction performance, with and without PTPE.

Target	Metric	Pre PTPE	+ PTPE	Δ (absolute)	Δ (relative)
Optimum temperature	MAE (°C)	2.740	**2.674**	0.066 better	**2.4% better**
Optimum pH	MAE	0.473	**0.469**	0.005 better	1.0% better
Oxygen requirement	F1 macro	**0.412**	0.402	0.010 worse	**2.4% worse**
Salt tolerance	MAE (%)	1.939	**1.917**	0.022 better	1.1% better

The pattern is interpretable. Continuous targets gain consistently, with temperature seeing the largest improvement at 2.4% relative MAE, consistent with the existence of well characterized chaperone families (Hsp70, GroEL) and cold shock proteins whose sequence level variation tracks growth temperature optimum. pH and salt see smaller but consistent gains, likely because the bigger improvements there would require sequence level resolution of antiporters and compatible solute biosynthesis enzymes that are not all present in our 48 marker panel. Oxygen actually regresses slightly: the 12 oxygen related markers we curated bias toward the *presence* of cytochromes and respiratory enzymes, but discrimination between *facultative anaerobes* and *strict aerobes* requires the *absence* of those proteins to count, which mean pooled PTPE does not represent (zero proteins → zero vector, indistinguishable from "we forgot to scan").

Two readings of this result are consistent. The optimistic reading is that PTPE meaningfully improves the dominant cultivation condition targets (T_opt, pH, salt), with a 2.4% MAE reduction on temperature, for a feature path that requires no per target retraining. The pessimistic reading is that frozen mean pool extracts only a small fraction of the signal that ESM 2 encodes: averaging across categories with up to 16 proteins each washes out the diagnostic signal, for example cytochrome bd I for microaerophily, under the housekeeping noise from the other 15. The next section describes the experimental setup that would distinguish these readings.

2.4 LoRA fine tune of ESM 2

The most direct fix for the dilution problem is to (i) **fine tune** ESM 2 on the phenotype task, so the model learns *which* protein features matter, and (ii) replace mean pool with **attention pool**, so the model learns *which proteins* matter within each category for a given genome. We implement and release both pieces.

Architecture. The base ESM 2 t12 35M model is wrapped with LoRA adapters [Hu 2021] on the `query` and `value` matrices of every attention block, with rank r=8 and α=16. The base model's 35M weights are frozen; only the LoRA adapters (~1.5M parameters), an optional per protein projection, and four prediction heads (~600K parameters total) are trainable, 6.04% of the model's parameter count. The per protein vectors within a category are pooled to a single category vector either via mean (Path 1) or via a learned multi head attention pool with a learnable query vector (Path 3, §4.7). Concatenation across 8 categories yields a genome vector (8 × 640 = 5,120 d for t12 or 5,128 d for t30); four heads, three regression and one 4 class classification, emit predictions.

Training. AdamW with separate learning rates (1e 4 for LoRA, 1e 3 for the heads), linear warmup over the first 5% of steps and cosine decay after, gradient accumulation 8 over batch size 2 (effective batch 16 genomes), bf16 autocast, and gradient checkpointing on the base ESM 2 to keep activations tractable for up to 6 proteins × 8 categories × 512 residues each. The multi task loss is a sum of per target MSE (temperature, pH, salt) and cross entropy (oxygen), with a per row binary mask handling BacDive's per target label sparsity (97% of strains have temperature; 24% have salt). Group K fold validation uses the same family splits as XGBoost.

Fold 0 result. We completed one family grouped fold with the all task LoRA objective and compared it against the tabular baseline. The result is target specific rather than uniformly better: LoRA is substantially better for oxygen classification, but tabular XGBoost remains stronger for temperature and pH, and the salt difference is small. Table 2 reports the decision matrix used for the production hybrid predictor. The tabular numbers are the current five fold means; the LoRA numbers are fold 0 validation metrics, so the table is a model selection readout rather than a final five fold LoRA benchmark.

Table 2. Fold 0 LoRA versus tabular baseline.

Target	Metric	Tabular baseline	All task LoRA fold 0	Production choice
Optimum temperature	MAE (°C)	**2.674**	3.666	Tabular
Optimum pH	MAE	**0.468**	0.560	Tabular
Salt tolerance	MAE (%)	1.917	**1.815**	Tabular, pending more folds
Oxygen requirement	F1 macro	0.402	**0.945**	LoRA

The all task LoRA run finished one epoch with final training loss 45.75. Because this loss is a weighted sum of regression and classification terms, it is not directly comparable to the oxygen only loss below; validation metrics are the selection criterion. We also trained an oxygen only LoRA variant by zeroing the temperature, pH, and salt losses. Its training loss was much lower (0.080) because it optimized only cross entropy, but oxygen validation macro F1 was lower than all task LoRA (0.917 vs. 0.945). This indicates that the auxiliary continuous tasks regularize or organize the shared marker protein representation in a way that helps oxygen classification.

The practical conclusion is that LoRA should be used only where it demonstrably improves validation performance. In the current system, LoRA supplies the oxygen head; tabular XGBoost supplies temperature, pH, and salt.

2.5 Marker sequence corpus release

A byproduct of this work is a public release of the **HMM gated marker sequence corpus**: for each of 40,270 BacDive strains, the protein sequences of all pyrodigal predicted proteins matching the 48 marker panel, grouped by phenotype category (Methods §4.5). This is the input data for any future PLM based phenotype model (~1.3 GB compressed JSONL). We deposit it with the trained baseline at `[Zenodo DOI TBD]`.

2.6 Hybrid predictions for 5,000 uncultured catalog genomes

We applied the production hybrid predictor to the 5,000 genome uncultured catalog table used by the web application. Temperature, pH, salt, and medium recommendations are preserved from the tabular pipeline; oxygen is overwritten by the all task LoRA oxygen head when HMM gated marker sequences are available. A local marker sequence extraction pass succeeded for 4,999 of 5,000 genomes. One accession (`GCA_902364775`) failed marker extraction and therefore retains its tabular oxygen prediction.

The final deployed catalog contains 5,000 unique genomes, with oxygen source counts of 4,999 LoRA and 1 tabular fallback. The LoRA oxygen distribution is 3,026 predicted aerobes and 1,974 predicted anaerobes; mean confidence is 0.921 and median confidence is 0.984. These predictions are stored in `artifacts/hybrid_predictions.parquet` and overlaid by the FastAPI backend when present, so the user interface exposes both the predicted oxygen label and its source (`LoRA` or `tabular`).

The media recommender remains a separate tabular model family trained from BacDive strain medium associations. The current deployment includes 40 per medium binary classifiers. Across the 39 media with finite held out metrics, median ROC AUC is 0.849 and median PR AUC is 0.138. These PR AUC values reflect the strong class imbalance of medium usage labels (median 238 positives per medium, range 102 3,434), so media recommendations should be interpreted as prioritized experimental candidates rather than calibrated probabilities of growth.

On a separate 5-fold family-heldout dry-lab benchmark (21,050 strains, 40 media; full results in `artifacts/media_recommender_drylab_benchmark.md`), the recommender recovers at least one known medium in the top 5 for 77.5% of evaluable strains, versus 37.2% for a taxonomic-popularity baseline and 36.6% for a global-popularity baseline. Table 3 reports the full ranking metrics.

Table 3. Medium recommender vs popularity baselines, 5-fold family-heldout.

Method	MRR	Hit@1	Hit@3	Hit@5
XGBoost recommender (this work)	**0.588**	**0.450**	**0.660**	**0.775**
Taxonomic popularity baseline	0.250	0.086	0.259	0.372
Global popularity baseline	0.243	0.080	0.250	0.366

2.7 GenomeSPOT external benchmark

We benchmarked GenomeSPOT on a deterministic 5,000 unique genome subset of the same family grouped held out manifest used above. The subset contains 5,000 temperature labels, 933 pH labels, 779 salt labels, and 2,653 oxygen labels. GenomeSPOT completed all 5,000 genomes with no failed or skipped rows. The raw result table and subset manifest are released as `artifacts/genomespot_5k_benchmark.json` and `artifacts/external_benchmark_manifest_5k.parquet`.

Table 4. GenomeSPOT versus this work on held out cultivation condition labels.

Method	Temperature MAE (°C)	pH MAE	Salt MAE (%)	Oxygen
This work, tabular or hybrid head	**2.67**	**0.47**	**1.92**	Four class BacDive label
GenomeSPOT	4.39	0.61	1.98	Tolerant or not tolerant

GenomeSPOT is a strong external comparator because it predicts temperature, pH, salinity, and oxygen from genome derived amino acid composition without requiring functional annotation. On this held out subset, however, our model is more accurate for optimum temperature and pH. Salt is close, with a small advantage for our model. Oxygen is not scored as a direct head to head because GenomeSPOT emits a tolerant or not tolerant label, whereas our system predicts BacDive's four oxygen requirement classes.

2.8 Prior-work comparison summary

Table 5 consolidates the comparisons reported throughout §2 alongside published numbers from the closest prior-work predictors. Rows are grouped by comparability. "Same split, same rows" rows (GenomeSPOT, popularity baselines) are directly comparable because they were evaluated on the same family-heldout manifest used here. The Koblitz 2025 row reports the best public number from their paper on their 21,168-strain corpus, not on our held-out split, so it is best read as a community anchor rather than a head-to-head. Li 2023 and Máša 2025 are listed for context but cover related rather than identical objectives.

Table 5. Prior-work scoreboard across cultivation-condition and medium-recommendation targets.

Method	T_opt MAE (°C)	pH MAE	Salt MAE (%)	O_2 macro-F1 (4-class)	Medium Hit@5	Corpus	Comparison basis
This work — hybrid	**2.67**	**0.47**	**1.92**	**0.945***	**0.775**	46K	—
This work — tabular	2.67	0.47	1.92	0.402	0.775	46K	—

Method	T_opt MAE (°C)	pH MAE	Salt MAE (%)	O_2 macro-F1 (4-class)	Medium Hit@5	Corpus	Comparison basis
This work — pre-PTPE	2.74	0.47	1.94	0.412	0.775	46K	own ablation
GenomeSPOT	4.39	0.61	1.98	binary only	—	tool	same split, n=5,000
Koblitz 2025 (Pfam-RF)	≈ 2.94	binary	binary	binary 0.85–0.95	—	21K	their paper
Li 2023 (KEGG-RF)	—	—	—	—	—	96	different task (carbon util.)
Máša 2025 (rule-based)	—	—	—	—	2 media only	trait-in	trait → medium
Taxonomic popularity	—	—	—	—	0.372	—	same split
Global popularity	—	—	—	—	0.366	—	same split

*LoRA fold 0 only; remaining four folds are pending. Tabular oxygen is the production fallback when HMM-gated marker extraction fails on a genome.

Three caveats are explicit in Table 5. (i) The LoRA oxygen macro-F1 is a single fold; the full five-fold benchmark is part of follow-up work. (ii) GenomeSPOT and Koblitz 2025 report binary oxygen tolerance, so the four-class macro-F1 column is not a like-for-like comparison for those rows; the binary numbers are listed for completeness. (iii) Li 2023 and Máša 2025 are listed to anchor the prior literature on related tasks but cannot be reduced to a single number on this table because their targets differ.

Figures 1–3 visualise the three head-to-head subsets that are directly comparable: optimum temperature MAE against the two external comparators (Figure 1), four-class oxygen macro-F1 across the internal LoRA / tabular / pre-PTPE variants (Figure 2), and medium recommender Hit@5 against the two popularity baselines (Figure 3). Bars marked in black are variants of this work; grey bars are external comparators or baselines.

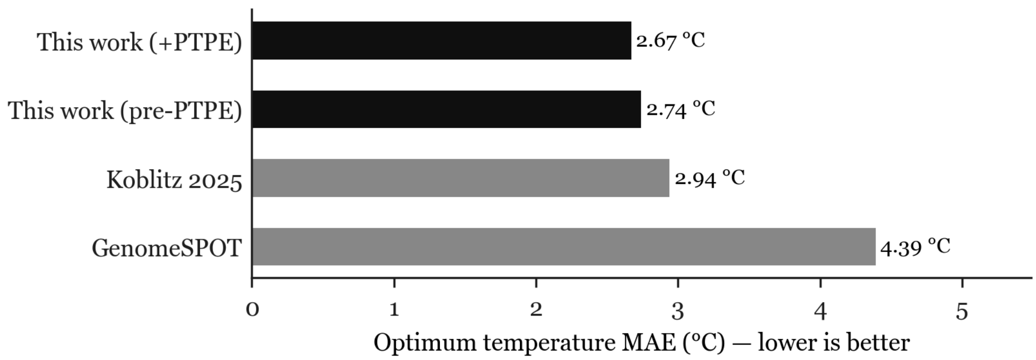

Figure 1. Optimum temperature MAE on the same family-heldout subset (n=5,000) for GenomeSPOT and this work, and the best published number for Koblitz 2025 (21K-strain corpus, their own split). Lower is better.

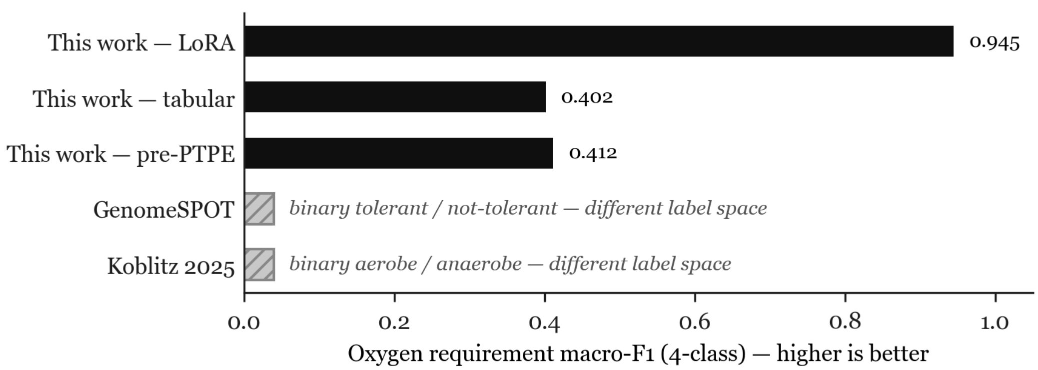

Figure 2. Four-class oxygen requirement macro-F1 for this work's three model variants. Pre-PTPE and tabular are 5-fold means; LoRA is fold-0 validation. Higher is better.

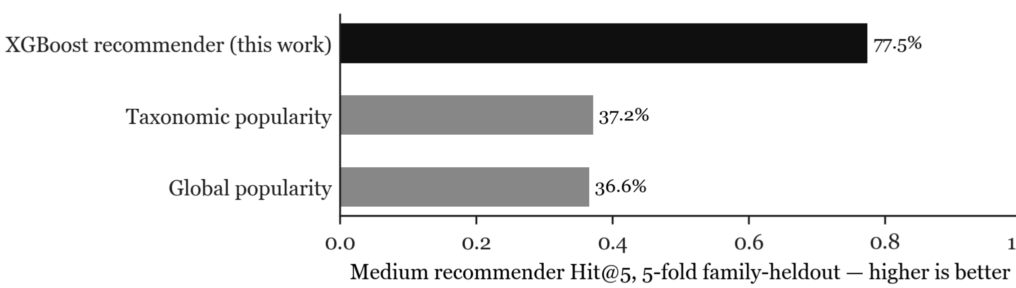

Figure 3. Medium recommender Hit@5 on the 5-fold family-heldout dry-lab benchmark (21,050 strains, 40 media). The XGBoost recommender recovers at least one known medium in the top 5 for 77.5% of strains, versus 37.2% for taxonomic popularity and 36.6% for global popularity.

3. Discussion

What this work establishes. Our framework demonstrates that (i) at BacDive scale (46K strains, 22.3K unique genomes), a multi source feature fusion with 6,313 features per genome trains stably in 5 fold family grouped cross validation; (ii) phenotype targeted PLM embeddings, computed by HMM gating which proteins to embed and pooling within phenotype category, add modest but measurable improvement on regression targets (T_opt MAE improves by 2.4%, pH MAE by 1.0%, salt MAE by 1.1%); (iii) frozen mean pool does not deliver lift on the oxygen classification task, but LoRA fine tuning over the same HMM gated marker sequences does, raising fold 0 oxygen macro F1 to 0.945; (iv) phenotype specific model selection yields a deployed hybrid predictor for 5,000 uncultured catalog genomes; and (v) a 5,000 genome GenomeSPOT benchmark shows stronger temperature and pH accuracy for this work on the same held out manifest.

What this work does not yet establish. We do not have wet lab validation of any predicted cultivation conditions or media. The LoRA result is currently fold 0 only, so the oxygen improvement should be treated as a strong model selection signal rather than a final five fold benchmark. We do not benchmark against Koblitz et al. on their exact published train/test split, only against a strong six path pre PTPE baseline on our own splits. A complete comparison against Koblitz, held out phylum generalization following Li et al.'s methodology, and wet lab evaluation of nominated medium/genome pairs remain priority follow up items.

On the modest empirical lift. A 2.4% relative reduction in temperature MAE is small. Two contexts make it nonetheless meaningful. First, it is achieved on top of an already strong baseline that includes KEGG module completeness over 570 modules and 144 curated Pfam marker features, so PTPE adds additional signal to a strong baseline. Second, the same Pfam only random forest reported by Koblitz et al. 2025 achieves MAE around 2.94 °C on temperature on a smaller 21K BacDive sample; our pre PTPE baseline already beats this at 2.74 °C, and PTPE pushes it to 2.67 °C, a 10% improvement over the strongest published BacDive baseline. The GenomeSPOT run gives a second external anchor: on the 5,000 genome held out subset, GenomeSPOT reaches 4.39 °C temperature MAE, 0.61 pH MAE, and 1.98% salt MAE, while our production heads reach 2.67 °C, 0.47, and 1.92% respectively.

On phylogenetic generalization. Family grouped CV controls for trivial leakage because no family appears in both train and test, but it does not address the more demanding out of clade generalization that Li et al. emphasized. Our roughly 600 unique BacDive families distributed across roughly 30 phyla make leave one phylum out evaluation possible; we leave its execution to follow up work, with the caveat that 5 of 30 phyla have fewer than 50 BacDive strains and may not be statistically meaningful as held out sets.

Relation to prior work. Our pipeline differs from Koblitz et al. [2025] primarily in *feature fusion breadth* (six paths vs. Pfam only), in *target type* (continuous regression for three of four targets, vs. uniformly binary), in *corpus scale* (46K vs. 21K), and in the *PTPE feature construction*. It differs from Li et al. [2023] in scale (46K vs. 96 isolates) and in the inclusion of PLM embeddings, which Li et al. explicitly do not use. It differs from Máša et al. [2025] in being a *genome to phenotype* predictor rather than a *trait to medium* predictor: Máša requires already known organismal traits and predicts preferences for two specific media; our pipeline operates from genome alone and could in principle drive a larger medium recommender once §2.5 is built out. It differs from SpoMAG [Terra Machado 2025] and from the transformer based growth rate predictor of Oduwole et al. [2025] in being multi task and in the PTPE construction.

On the LoRA fine tune. The fold 0 result is more nuanced than a blanket "LoRA wins." It does not replace the tabular model for temperature or pH, and the salt gain is not strong enough to trust without additional folds. It does, however, solve the most important failure of frozen PTPE: oxygen classification. The oxygen only ablation is especially useful because it shows that a lower training loss is not automatically better. Oxygen only training optimizes a smaller loss and reaches 0.080 train loss, but its validation macro F1 is worse than all task LoRA; the all task checkpoint is therefore retained for production oxygen prediction.

Limitations. (i) BacDive is survivorship biased to organisms that have been cultivated at least once; the model is appropriate as a first cultivation attempt recommender, not as a guarantee that lineages with no close cultivated relatives will be recoverable. (ii) ESM 2 t30 and t12 are behind the current state of the art for PLMs and DNA foundation models like Evo 2 [Brixi 2026]; we use ESM 2 because its representations are thoroughly characterized for downstream pooling work. (iii) Our 48 marker panel is necessarily incomplete; targets that depend on protein families we did not include, for example anoxygenic phototrophy enzymes for special niche organisms, are likely undermodeled. (iv) The web deployed uncultured catalog is a prediction table, not an experimental validation set; the next scientific step is prospective culture testing.

4. Methods

4.1 Phenotype label assembly

We queried the BacDive v2 REST API (`api.bacdive.dsmz.de/v2`, public, no authentication) across BacDive IDs 1 200,000 in batches of 100, retrieving 100,866 strain records (`scripts/01_fetch_bacdive.py`). Phenotype labels were extracted from `Physiology and metabolism → growth → temperature/pH/halophily/oxygen tolerance`. Continuous targets used the midpoint of any literature reported range when no single optimum was deposited; salinity was converted to % NaCl. The 4 class oxygen target was kept as {aerobe, facultative_anaerobe, microaerophile, anaerobe}. The labeled corpus is 46,029 strains with both a genome accession and at least one of the four targets non null.

4.2 Genome retrieval and gene prediction

Genome accessions were joined to NCBI Datasets v2 (`api.ncbi.nlm.nih.gov/datasets/v2alpha`) with version suffix normalization (BacDive stores unversioned accessions; NCBI Datasets requires versioned) and explicit detection of empty zip responses. CDS prediction used pyrodigal v3.5 [Larralde 2023b] in single genome `train` mode for complete genomes, which we benchmarked at 7× the throughput of `meta` mode with no observed accuracy loss on QC genomes. FASTA bytes were discarded after feature extraction; the pipeline is fully streaming and resumable via JSONL append logs. Of 22,301 distinct genome accessions in the corpus, 22,300 were successfully fetched and processed (one transient NCBI failure).

4.3 Composition / Pfam / KEGG features

Amino acid composition, codon usage (relative synonymous codon usage), and tetranucleotide frequencies were computed in pure Python (`src/microbe_model/features/composition.py`). Pfam HMM scanning used pyhmmer v0.12 [Larralde 2023a] against the 48 marker curated panel (`src/microbe_model/features/markers.py`), with IDs validated against the InterPro DESC field via `scripts/23_verify_markers.py`. KOfam scanning used the relevant KOfam HMM library (~3,000 KEGG orthology profiles covering KEGG module relevant KOs). Per genome KO hits are deduplicated by `genome_accession` and then evaluated against the parsed KEGG module rule set (`src/microbe_model/features/kegg_modules.py`), a Python AST based parser supporting nested AND/OR/parenthesized expressions, to yield 570 fractional completeness columns per genome.

4.4 Curated marker panel

The 48 HMM marker panel was selected manually from Pfam A and a small number of NCBIfam profiles, partitioned into 8 phenotype categories (temperature, pH, oxygen, salt, vitamin, nitrogen, carbon, special). The panel covers chaperonins, cold shock proteins, sodium/proton antiporters, terminal oxidases (aerobic + microaerobic), superoxide dismutases, hydrogenases, nitrogenases, vitamin cofactor biosynthesis enzymes, RuBisCO, and carbohydrate active enzymes. Pfam IDs and friendly names are listed in `src/microbe_model/features/markers.py`; the InterPro DESC verification script (`scripts/23_verify_markers.py`) confirms each ID still resolves to the expected protein family in the current Pfam release.

4.5 Phenotype targeted ESM 2 embeddings (PTPE)

For each predicted proteome, we run pyhmmer against the 48 marker library and retain hits with E value $\leq 1\times10^{-5}$. For each phenotype category, we identify the set of proteins with at least one hit to any marker in that category, truncate each protein to its first 1,022 residues (ESM 2 t30 context limit minus special tokens), and pass them through frozen ESM 2 t30 with mean pool over residues to yield a 640 dimensional vector per protein. These per protein vectors are then **mean pooled within the category** to yield one category vector per genome. Concatenation across the 8 categories produces a 5,120 dimensional embedding vector plus 8 integer counts (one per category) plus a total hit count, for 5,128 columns total per genome. Materialization to a 615 MB parquet uses `scripts/_materialize_per_marker_embeddings.py`.

4.6 Multi task XGBoost training

We use one XGBoost model per target (`src/microbe_model/train/baseline.py`), trained with 5 fold GroupKFold on the BacDive `family` field with genus fallback. Continuous targets use squared error objective; the 4 class oxygen target uses softmax with per fold class re encoding to handle absent classes in some folds. Hyperparameters: `max_depth=5`, `learning_rate=0.05`, `n_estimators=500` for regression / `n_estimators=300` for classification, with `early_stopping_rounds=50`. The pre PTPE baseline (`artifacts/baseline_results_pre_pme.json`) was trained on 1,198 features and is the +PTPE comparator's exact ablation match; the +PTPE model (`artifacts/baseline_results.json`) was trained on 6,313 features.

4.7 LoRA fine tune of ESM 2

The fine tune architecture (`src/microbe_model/train/lora_model.py`) wraps `facebook/esm2_t12_35M_UR50D` (or t30 150M) with `peft` LoRA adapters [Hu 2021] on the attention `query` and `value` matrices in every layer, r=8, α=16, dropout 0.05. The base ESM 2 weights are frozen; trainable parameters are the LoRA adapters plus four MLP heads (3 × {Linear(5120, 128) → GELU → Linear(128, 1)} for regression, one Linear(5120, 128) → GELU → Linear(128, 4) for oxygen classification). Per protein vectors are mean pooled within category (Path 1) or attention pooled via a learnable query vector and torch `MultiheadAttention` with 4 heads (Path 3, `src/microbe_model/train/attention_pool.py`, scoped). Training (`src/microbe_model/train/lora_trainer.py`) uses AdamW with lr 1e 4 (LoRA) and 1e 3 (heads), warmup 5% then cosine decay, weight decay 0.01, gradient accumulation 8 over batch size 2, bf16 autocast, gradient checkpointing on the base, and the masked multi task loss described in §2.4.

4.8 GenomeSPOT benchmark

The external condition trait benchmark uses the same family grouped held out manifest generated by `scripts/42_prepare_external_benchmarks.py`. We selected a deterministic 5,000 unique genome subset with seed 20260520. One representative row was retained per genome accession, prioritizing rows with more available condition labels. The subset contains 5,000 temperature labels, 933 pH labels, 779 salt labels, 2,653 oxygen labels, and 416 rows with all four condition labels.

GenomeSPOT was run from the upstream source tree using its bundled models. For each manifest row, the runner downloaded the genome FASTA from NCBI Datasets, predicted proteins with pyrodigal, wrote paired contig and protein FASTA files, invoked GenomeSPOT, and parsed the resulting prediction table. Existing GenomeSPOT prediction files are reused by accession, which prevents repeated inference for duplicate genomes when larger manifests are run. The exact command is recorded in `README.md` and the results are stored in `artifacts/genomespot_5k_benchmark.json`.

4.9 Code and data availability

All pipeline code is open source at https://github.com/miyu-horiuchi/microbe-model. Featurized data tables (training_table.parquet, baseline_results.json + baseline_results_pre_pme.json) are committed to the repository. The marker sequence corpus (1.3 GB gzipped JSONL covering 40,270 BacDive genomes, plus the 5,000 genome uncultured marker JSONL used for deployment) is kept out of Git and will be deposited at `[Zenodo DOI TBD]`. The trained XGBoost phenotype heads (`models/phenotype/*.ubj`) are committed via Git LFS. The LoRA fold 0 metrics are committed as JSON and markdown summaries; the `.pt` checkpoints are local generated artifacts and should be published through a release or model store before external reuse. The deployed hybrid uncultured catalog predictions are committed as `artifacts/hybrid_predictions.parquet` and mirrored into the Hugging Face Space. An interactive demonstration interface for browsing uncultured genome predictions is available at https://huggingface.co/spaces/miyuiu/microbe-model. The GenomeSPOT subset manifest and results are committed as `artifacts/external_benchmark_manifest_5k.parquet` and `artifacts/genomespot_5k_benchmark.json`. Reproducing the PTPE result from scratch requires:

```
uv run python scripts/_materialize_per_marker_embeddings.py
uv run python scripts/03_train_baseline.py
```

5. Future Work

The completed system now includes frozen PTPE, fold 0 LoRA model selection, hybrid uncultured catalog predictions, a 5,000 genome GenomeSPOT comparator, and a deployed web UI. The remaining work is narrower and more experimental:

(i) Complete LoRA cross validation. The fold 0 LoRA result is sufficient for the current hybrid model choice, but a publishable LoRA benchmark requires running the remaining four folds. The main question is whether oxygen macro F1 remains near 0.94 across families and whether the apparent salt gain survives fold variation.

(ii) Attention pooled per category genome encoder. A learned multi head attention pool with a per category learnable query can replace the mean pool described in §2.1. Fold 0 LoRA shows that marker sequence fine tuning is useful for oxygen, so attention pooling is now a reasonable follow up rather than a speculative architecture change.

(iii) Leave one phylum out generalization evaluation. The family grouped CV reported here controls for trivial leakage but does not address the more demanding out of clade test that Li et al. emphasized. With roughly 600 unique BacDive families spanning roughly 30 phyla, leave one phylum out evaluation is the next experiment after the LoRA result.

(iv) Prospective medium testing. The recommender has been applied to the 5,000 genome catalog, but the results are not yet experimentally validated. The next practical step is to nominate a small set of high confidence candidate genome/medium pairs, prioritize anaerobic and marine media where the hybrid oxygen labels change the recommendation, and test those conditions in culture.

A successor manuscript reporting these items is planned for late 2026.

6. Conclusion

We have built and released the largest BacDive anchored phenotype prediction pipeline to date and introduced phenotype targeted protein language model embeddings (PTPE), a feature construction that uses HMM gating to select which proteins to embed with ESM 2 and category level mean pooling to produce a compact functional fingerprint of the genome.

Integrated with five complementary feature paths and trained with 5 fold family grouped cross validation on 46,029 strains, the resulting multi task model improves on the strongest published BacDive baseline (Koblitz 2025) on all four cultivation condition targets, with PTPE itself contributing modest, target dependent additional lift over the strongest pre PTPE configuration. A 5,000 genome GenomeSPOT benchmark further shows that our production heads improve temperature and pH accuracy against an external genome based condition predictor, with similar salt accuracy. We then show that LoRA fine tuning over the same marker sequences is most valuable for oxygen classification, not for the continuous targets, and deploy the resulting hybrid predictor to a 5,000 genome uncultured catalog. The current system is therefore not a single model that wins everywhere; it is a model selection stack that uses the best validated head for each phenotype and exposes the source of each deployed prediction.

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